

Scientific Calculator Project Report

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1 Introduction

This project implements a scientific calculator using the Arduino Uno. It supports basic arithmetic, trigonometric (and inverse) functions, logarithmic/exponential operations, and complex expressions with proper operator precedence. A JHD162A LCD display shows the results, while push buttons serve as input. A recursive descent parser evaluates expressions accurately and efficiently.

2 Aim

- Design a calculator that performs:
 - Arithmetic: $+$, $-$, $*$, $/$, $^$
 - Trigonometry: $\sin$, $\cos$, $\tan$ (in degrees)
 - Inverse Trigonometry: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ (in degrees)
 - Log/Exp: $\ln$, $\log$, $\exp$
 - Others: $\sqrt{x}$, abs , $\%$
- Evaluate expressions using a recursive descent parser with proper precedence and parentheses.
- Provide real-time feedback via a JHD162A LCD display.
- Optimize resource use on the Arduino Uno.

3 Components Required

- Arduino Uno
- JHD162A LCD display
- 14 Push buttons:
 - 10 for digits (0–9)
 - Clear (C) and Enter (=)
 - 2 Shift buttons: Arithmetic Shift (Shift-A) and Scientific Shift (Shift-S)
- Wires, breadboard

4 Hardware Configuration

4.1 Pin Connections

4.1.1 LCD Display (4-bit Mode)

LCD Pin	Function	Arduino Pin	Notes
RS	Register Select	D8 (PB0)	LOW: Command; HIGH: Data
E	Enable	D9 (PB1)	Latches data
D4	Data Bit 4	D10 (PB2)	4-bit interface
D5	Data Bit 5	D11 (PB3)	4-bit interface
D6	Data Bit 6	D12 (PB4)	4-bit interface
D7	Data Bit 7	D13 (PB5)	4-bit interface

Additional connections: VSS (GND), VDD (+5V), VO (potentiometer for contrast), RW (GND), A (+5V backlight), K (GND).

4.1.2 Push Button Connections

Button	Function	Arduino Pin	Notes
Digits 0–5	Enter digits 0–5	D2, D3, D4, D5, D6, D7	Active-low; internal pull-ups
Digits 6–9	Enter digits 6–9	A0, A1, A2, A3	Digital inputs; pull-ups enabled
Clear (C)	Clear input	A4	Resets expression
Enter (=)	Evaluate expression	A5	Triggers computation
Extra Button 1	Additional function	(Assign as needed)	
Extra Button 2	Additional function	(Assign as needed)	

Note: The extra two push buttons can be used for further functions or for shift modes if needed.

4.1.3 Shift Button Connections

Button	Function	Arduino Pin	Notes
Arithmetic Shift (Shift-A)	Toggle arithmetic mode	D0 (PD0)	Repurposed from serial RX
Scientific Shift (Shift-S)	Toggle scientific mode	D1 (PD1)	Repurposed from serial TX

4.1.4 Power Supply

Supply	Connection
VCC	+5V
GND	Ground

4.2 Notes

- Internal pull-ups are enabled (using `digitalWrite(pin, HIGH)`).
- Buttons are active-low (pressed = LOW).

- A 30 ms debounce delay is applied.

5 Software Design

5.1 Overall Architecture

The software is modular, divided into:

- Input Handling:** Scans push buttons, applies a 30 ms debounce delay, and detects mode switches.
- User Interface (UI):** Updates the LCD with the current expression, mode menus, and results. It initializes the LCD (4-bit mode), clears the screen, and sets the cursor.
- Expression Evaluation:** Implements a recursive descent parser that enforces operator precedence, handles parentheses, and supports function calls.

5.2 Input Handling

- **Button Scanning:** Uses `digitalRead()` to poll each push button.
- **Debouncing:** A 30 ms delay is applied to prevent false triggers.
- **Mode Switching:** Shift-A and Shift-S buttons toggle between arithmetic and scientific modes.

5.3 User Interface (UI)

- **LCD Initialization:** Functions like `lcd_init()`, `lcd.clear()`, and `lcd.setCursor()` prepare the display.
- **Dynamic Display:** The current expression and mode menus are shown in real time. Error messages and results (to 4 decimal places) are displayed as needed.

5.4 Expression Evaluation (Recursive Descent Parser)

- **Operator Precedence:**
 1. Level 1: + and -
 2. Level 2: *, /, and %
 3. Level 3: ^ (right-associative)
- **Parentheses:** Extraneous outer parentheses are removed.
- **Function Support:** Recognizes and evaluates:
 - Trigonometric: `sin`, `cos`, `tan` (in degrees)
 - Inverse Trigonometric: `asin`, `acos`, `atan` (in degrees)
 - Log/Exp: `exp`, `ln`, `log`
 - Others: `sqrt`, `abs`
- **Numerical Methods:** Utilizes Euler's method, Newton-Raphson, and Riemann sums for function approximations.

5.5 Shift Modes

- **Arithmetic Shift Mode (Shift-A):**
 - **Navigation:** Use buttons 8 (Next) and 9 (Prev) to switch pages.
 - **Mapping:**
 - * Page 0: +, -, *, /
 - * Page 1: ^, %, (,)
 - * Page 2: Backspace (BS)
- **Scientific Shift Mode (Shift-S):**
 - **Navigation:** Use buttons 8 and 9 for page control.
 - **Mapping:**
 - * Page 0: sin, cos, tan, exp
 - * Page 1: ln, sqrt, log, abs
 - * Page 2: asin, acos, atan

The LCD displays the current mode and page for user guidance.

5.6 Error Handling

- Checks for division by zero, mismatched parentheses, and domain errors (e.g., sqrt of a negative number).
- Invalid selections in shift modes trigger brief error messages.

6 Operation Workflow

1. **Startup:** The LCD shows "Simple Calc" for 2 seconds, then displays "Enter:".
2. **Normal Mode:** Users input digits and operators; pressing = evaluates the expression.
3. **Shift Modes:** Toggled by Shift-A (D0) or Shift-S (D1); navigation with buttons 8/9; selection with buttons 0–3.
4. **Result:** Displayed to 4 decimal places; new input clears the result.

7 Performance Specifications

- **Voltage:** 5V DC
- **Current:** ~50 mA (active), <10 mA (idle)
- **Precision:** IEEE 754 emulation; results to 4 decimal places.
- **Expression Limit:** 64 characters.
- **Speed:** Arithmetic <10 ms; trigonometric ~200 ms.

8 Code Structure

The modular code includes functions for LCD control, input scanning, expression evaluation, and shift mode management. The main firmware is contained in `main.c`.